

SQL CONSTRAINTS

Implementation guide, constraint validations, and database engine error responses.

PRIMARY KEY

Unique + Non-Null

FOREIGN KEY

Referential Integrity

NOT NULL

Mandatory Value

CHECK

Domain Validation

Task 1: Table Creation with PRIMARY KEY

PLAYLIST

Create a table named Playlist with song_id as the primary key column.

```
CREATE TABLE Playlist (  
  song_id INT PRIMARY KEY,  
  song_name VARCHAR(100),  
  artist VARCHAR(100)  
);
```

Execution & Example Insertion:

```
INSERT INTO Playlist (song_id, song_name, artist)  
VALUES (1, 'Bohemian Rhapsody', 'Queen');
```

Query OK, 1 row affected (0.01 sec)

Task 2: Table Creation with UNIQUE, NOT NULL & DEFAULT

USERPROFILE

Create UserProfile with multiple field constraints including a default city location.

```
CREATE TABLE UserProfile (  
  user_id INT PRIMARY KEY,  
  username VARCHAR(50) UNIQUE,  
  email VARCHAR(100) NOT NULL,  
  city VARCHAR(50) DEFAULT 'Ahmedabad'  
);
```

Example Insertion (Omitting City to demonstrate DEFAULT):

```
INSERT INTO UserProfile (user_id, username, email)  
VALUES (101, 'aaron_dev', 'aaron@example.com');
```

Query OK, 1 row affected. Column 'city' populated with default value 'Ahmedabad'.

Task 3: Instagram-Style Follows with Composite FOREIGN KEYS

FOLLOWS

Create Follows referencing UserProfile(user_id) for both follower and followed relationships.

```
CREATE TABLE Follows (  
    follower_id INT,  
    followed_id INT,  
    PRIMARY KEY (follower_id, followed_id),  
    FOREIGN KEY (follower_id) REFERENCES UserProfile(user_id),  
    FOREIGN KEY (followed_id) REFERENCES UserProfile(user_id)  
);
```

Example Insertion:

```
-- Requires user_id 101 and 102 to exist in UserProfile  
INSERT INTO Follows (follower_id, followed_id) VALUES (101, 102);
```

Query OK, 1 row affected (0.01 sec)

Task 4: NOT NULL Constraint Violation Test

EXCEPTION TEST

Attempting to insert a row into UserProfile with a NULL email value:

```
INSERT INTO UserProfile (user_id, username, email, city)  
VALUES (102, 'johndoe', NULL, 'Mumbai');
```

POSTGRESQL ERROR MESSAGE

ERROR: null value in column "email" violates not-null constraint
DETAIL: Failing row contains (102, johndoe, null, Mumbai).

MYSQL / MARIADB ERROR MESSAGE

ERROR 1048 (23000): Column 'email' cannot be null

Task 5: Adding and Violating a CHECK Constraint

VALIDATION TEST

Add a constraint requiring song_name length ≥ 3 , then insert a 2-character song name ('Hi'):

```
ALTER TABLE Playlist  
ADD CONSTRAINT check_song_name_length CHECK (LENGTH(song_name) >= 3);  
  
INSERT INTO Playlist (song_id, song_name, artist)  
VALUES (2, 'Hi', 'Anirudh');
```

POSTGRESQL ERROR MESSAGE

ERROR: new row for relation "playlist" violates check constraint "check_song_name_length"
DETAIL: Failing row contains (2, Hi, Anirudh).

MYSQL ERROR MESSAGE (V8.0.16+)

ERROR 3819 (HY000): Check constraint 'check_song_name_length' is violated.